

Exercises (1)

Exercise 1. Show that if $X \subseteq \mathbb{R}^n$ is a convex set and the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strictly convex, then the problem

$$\begin{aligned} \min f(x) \\ x \in X \end{aligned}$$

has a unique solution if it has any solutions.

Exercise 2. Let $X \subseteq \mathbb{R}^n$ be a closed and convex set and let $x \notin X$.

1. Show that the orthogonal projection of x on X ($\text{Proj}_X(x) := \arg \min_{u \in X} \|x - u\|$) exists and is unique.
2. Show that: $\forall x \notin X, \quad v^* = \text{Proj}_X(x) \Leftrightarrow \langle x - v^*, v - v^* \rangle \leq 0, \quad \forall v \in X$.
3. Show that: $\forall x \notin X, \forall y \notin X, \|\text{Proj}_X(x) - \text{Proj}_X(y)\| \leq \|x - y\|$.

Exercise 3. Recall that for a hyperplane $H = \{x \in \mathbb{R}^n : \langle s, x \rangle = b\}$, positive and negative halfspaces are defined as $H^+ = \{x \in \mathbb{R}^n : \langle s, x \rangle \geq b\}$ and $H^- = \{x \in \mathbb{R}^n : \langle s, x \rangle \leq b\}$. The interior of H_+ is defined as $\text{int}(H^+) = \{x \in \mathbb{R}^n : \langle s, x \rangle > b\}$. $\text{int}(H^-)$ is defined similarly.

Given a set X , a hyperplane H such that $X \subset H^+$ or $X \subset H^-$ is called a supporting hyperplane. Let X be a closed convex set and $x \notin X$.

1. Show that there exists a hyperplane H such that $X \subseteq H^-$ and $x \in \text{int}(H^+)$.
2. Show that $X = \bigcap_{X \subseteq H^-} H^-$, i.e., X is the intersection of all such halfspaces containing X .

Exercise 4. For each of the following functions determine whether it is convex, concave, quasiconvex, or quasiconcave.

1. $f(x) = e^x - 1$ on $\mathbb{R}$.
2. $f(x) = -\ln(x)$ on $\mathbb{R}_+^*$.
3. $f(x) = x^4 - x^2$ on $\mathbb{R}$.
4. $f(x) = x^p, (p \in \mathbb{N})$ on $\mathbb{R}$.
5. $f(x) = |x|^\beta, (\beta > 0)$ on $\mathbb{R}$.
6. $f(x, y) = xy$ on $\mathbb{R}_{++}^2$.
7. $f(x, y) = \frac{1}{xy}$ on $\mathbb{R}_{++}^2$.
8. $f(x, y) = \frac{x}{y}$ on $\mathbb{R}_{++}^2$.

9. $f(x, y) = \frac{x^2}{y}$ on $\mathbb{R}_{++}^2$.

Exercise 5. Farkas' lemma.

Show the following fundamental theorem of linear inequalities:

Given any vectors g and a_i , $i \in \mathcal{A}$, the set

$$\mathcal{S} = \{s \mid \langle g, s \rangle < 0 \text{ and } \langle a_i, s \rangle \geq 0 \ \forall i \in \mathcal{A}\}$$

is empty if and only if

$$g \in C = \left\{ \sum_{i \in \mathcal{A}} \alpha_i a_i \mid \alpha_i \geq 0 \text{ for all } i \in \mathcal{A} \right\}.$$

Exercise 6. Some operations that preserve convexity of functions

1. Show that if $f : \Omega \rightarrow \mathbb{R}$ is convex, then also $\tilde{f}(x) := f(Ax + b)$ (with $A \in \mathbb{R}^{n \times m}$) is convex on the domain $\tilde{\Omega} = \{x \in \mathbb{R}^m \mid Ax + b \in \Omega\}$.
2. Show that if $f : \Omega \rightarrow \mathbb{R}$ is convex and $g : \mathbb{R} \rightarrow \mathbb{R}$ is convex and monotonely increasing, then the function $g \circ f : \Omega \rightarrow \mathbb{R}$, $x \mapsto g(f(x))$ is also convex.
3. Show that the supremum over a set of convex functions $f_i(x), i \in I$ is convex:
 $f(x) = \sup_{i \in I} f_i(x)$ is convex.